27/8/2026 FACULTY OF COMPUTING

Research Ethics Review Form

All students undertaking research in the Faculty of Computing must fill in and submit this form to the respective coordinator / supervisor as instructed by the program head.

SECTION 1

1. Project/research type

☒	B.Sc	☐	M.Sc	☐	MPhil/PhD

2. Project/research details

Project ID (if any)	R26-DS-010
Research Title	Multimodal Computational Intelligence for Enhancing Caregiver Performance and Care Delivery

3. Student details

Student ID	Name	Email	Phone
IT22178640	Dayana Priyadharshani Kumar (Group Leader)	it22178640@my.sliit.lk	0764043970
IT22057488	Harishalinee Elangovan	it22057488@my.sliit.lk	0775073188
IT22294098	Vidursha Prabagaran	it22294098@my.sliit.lk	0779890702
IT22289384	Kaushalya Nagenthraraja	it22289384@my.sliit.lk	0764232082

4. Supervisor details

Supervisor Name	Prof. Samantha Thelijjagoda
Co-supervisor Name	Mr. Samadhi Rathnayake
External supervisor	Mr. Velu Sivakumar

SECTION 2: DATA COLLECTION AND STORAGE

Read the following table before answering the questions.

Data Source	Description	Data Collection Methods
Primary Data	Primary data collection happens when researchers obtain information directly from the original sources.	Survey/ Interviews/ Observation/ Focus groups/ New Images etc.
Secondary Data	Datasets already gathered from previous research. This data is usually already analyzed and put into context. Sources of secondary data can include:	Books / Past research papers/ Annual reports/ Newspapers / Reports / Websites/ Online Databases etc.

Provide details of all data required for your project.

Research Component	Data required	Amount of Data Required	Data Source (Primary /Secondary)	Collection Method
Personalized Stress-Risk Detection and Relational Network Analysis	Wearable physiological data, Bluetooth proximity data, EMA self-reported stress surveys	212 caregivers, ~10 weeks (TILES-2018)	Secondary	Public research dataset (TILES-2018, published online)
Edge-Enabled Skeletal Movement Analysis System for Real-Time Patient Observable Fall Risk Detection and Prediction	Skeleton/action video clips (public benchmarks) + in-house webcam recordings of simulated fall scenarios	578 public clips + 93 in-house clips	Secondary + Primary	Public datasets (NTU RGB+D, UR Fall Detection) + own video recording with informed consent
Context-Aware Intelligent System for Automated ADL Extraction and Real-Time Documentation from Caregiver Voice Logs	Caregiver speech/accent data (for ASR) + daily ADL activity records (medication, feeding, hygiene, mobility, symptoms) logged from a caregiver's home	~6,656 speech utterances (public) + ~2,300 ADL activity records	Secondary + Primary	Public dataset (Svarah) + observation of caregiving activities, recorded into an Excel sheet at a caregiver's home
Personalized Sign Language Tutoring System for Caregivers	Sign-language video clips	2,364 sequences, 59 sign classes	Secondary	Public datasets (WLASL, ASL Citizen)

5. **DATA COLLECTION: If you require primary data, do you intend to collect data from following groups?**

☐	Children or young people
☐	Pregnant women
☐	Individuals with cognitive disabilities
☐	People suffering from specific diseases
☐	Animals
☑	Certain group of people involved in an organization

6. **DATA COLLECTION: If you are collecting information related to people, in what form will be the data collected?**

☑	Individually identifiable	Data can directly lead to the identification of an individual or organization
☑	Re-identifiable*	Data indirectly leads to the identification of a person or organization (even through direct identifiers are removed)
☐	Non-Identifiable	Data does not lead to the identification of a specific individual

**Anonymous data from which the individual could be discovered*

7. **DATA COLLECTION: Does your project possibly cause/involve any of the following?**

☐	Physical stress/distress, or discomfort to participants
☐	Psychological/mental stress/distress, or discomfort to participants
☑	Collection of sensitive/confidential data of individuals
☐	Collection of confidential data of organizations
☐	Development of a product harmful for a particular group or society

8. **DATA COLLECTION: Explain the procedures you will be following to ensure privacy of participants or organizations during data collection, data analysis and publishing of results.**

☑	Before the data collection, participants will be given an information sheet to create awareness
☑	Data will be collected only from the participants who have given consent through a consent form
☐	Personally identifiable data will not be collected

☑	If identifiable data is present in the data set, data will be either anonymized or coded

Additional Measures: Informed consent was obtained from all volunteers involved in Component 2's in-house webcam recordings. Component 3's ADL activity data was recorded as caregiving activity logs (Excel sheet) at a caregiver's home; an information sheet and consent process will be confirmed and arranged for this data prior to publication, and coding/anonymisation (e.g., patient/participant codes) will be applied before any external use. All four public secondary datasets are used strictly under their respective published terms of use and licenses.

9. DATA STORAGE: If you are collecting information related to people, in what form will the data be stored?

☐	Individually identifiable	☐	Re-identifiable	☑	Non-Identifiable

10. DATA STORAGE: Where will data be stored and what measures will be taken to ensure security of data ?

☐	Paper-based data stored in locked filing cabinets within SLIIT.
☑	Data stored in hard-drives and computers will be protected by password and encryption.
☑	Data will be anonymized and will not be re-identifiable.
☑	After the completion of the project, all data will be permanently deleted from all student devices. Only a single copy will be securely stored within a repository in the institution.

Additional measures: Data will be stored on encrypted personal devices accessible only to the four group members. Component 2's webcam recordings and Component 3's ADL activity logs will be coded/pseudonymised (e.g., participant/patient codes) before storage and analysis, reducing re-identification risk. Access to raw, uncoded data will be restricted to the responsible component owner only.

11. How would you rate the overall risk associated with the research? (These risks may occur when the collected data is exposed to unauthorized parties, etc.")

☐	High Risk	☑	Low Risk	☐	Negligible Risk

12. Declaration of the participants.

I/we certify that the details given above are accurate and agree to adhere to the conditions of approval specified by the panel. Furthermore, I/we will inform the FoC Research Project coordinator **of** any deviation from the specified details.

Student ID	Name	Signature
IT22178640	Dayana Priyadharshani Kumar	
IT22057488	Harishalinee Elangovan	
IT22294098	Vidursha Prabagaran	
IT22289384	Kaushalya Nagenthraraja	

13. Approval of the Supervisor

Supervisor Recommendation (please select one option)	☐ **Ethics committee Approval NOT Required** ☐ **Ethics committee Approval Required** **If approval required, clearly specify the reasons:**	
Supervisor Name	**Supervisor Signature**	**Date**
Prof. Samantha Thelijjagoda		

**** Please note that if the supervisor has recommended that ethics committee approval required, student should fill the 'Ethics Review Form' of the SLIIT ERC and submit it to the faculty Ethics Sub-committee.**